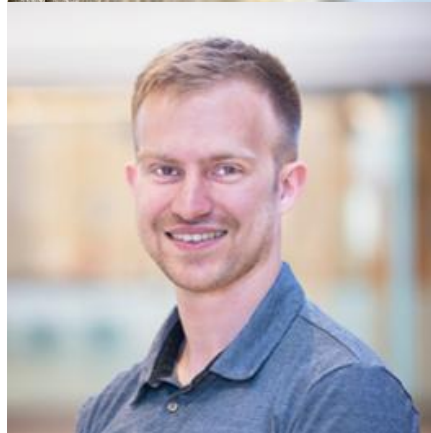


APSC 131 – Chemistry of Materials

Lecture 01 – Introduction

APSC 131 Course Coordinator



Peter Gilbert

Chemical Engineering

E-mail: APSC131instructor@queensu.ca

Office hours: By appointment

APSC 131 Course Instructors



Rachel Baker
Chemical Engineering



Scott Parent
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Hoang Dang
Chemical Engineering



Bradley Diak
Mechanical & Materials
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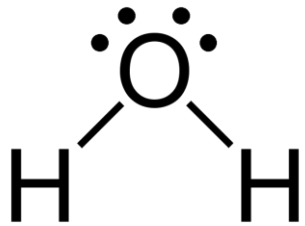
Mark Daymond
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What makes water... water?

Material Engineering Design

Material Structure



- Molar mass (M)
- Atomic number (Z)
- Atomic radius (r)
- Bonding
 - Ionic
 - Covalent
 - Metallic

Chemical & Physical Properties



- Density (ρ)
- Melting point (T_m)
- Heat capacity (C_p)
- Thermal conductivity (k)
- Electrical conductivity (σ)

Citation: medium.com

Engineering Properties



- Elasticity (E)
- Strength
- Ductility
- Hardness
- Toughness
- Corrosion resistance

Citation: evoke.pro

Processing



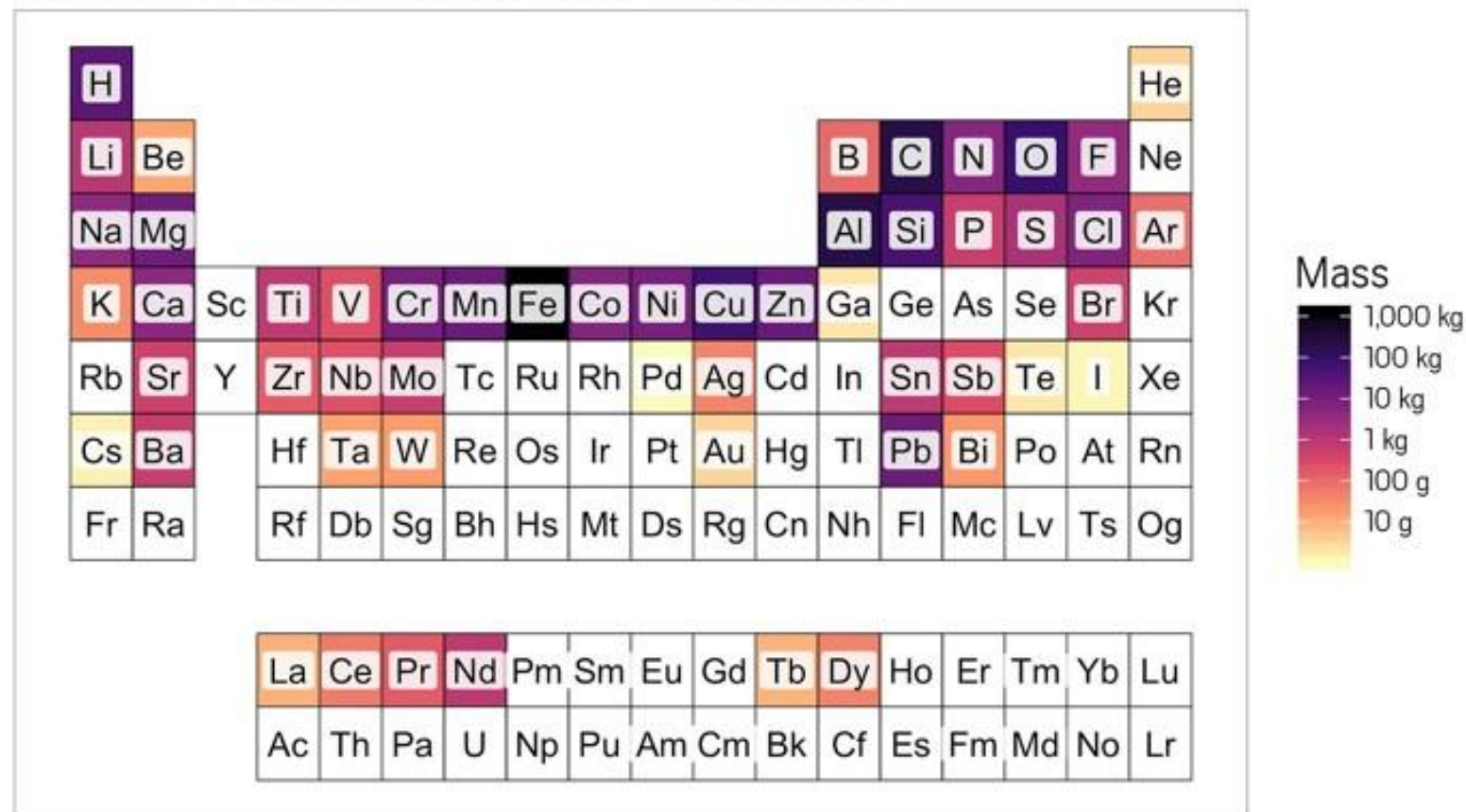
- Alloying
- Reinforcement
- Heat treatment
- Surface treatment

DOI: [10.3389/fpsyg.2021.638703](https://doi.org/10.3389/fpsyg.2021.638703)

Elements of Engineering

Every element is important, but some elements are more commonly encountered by engineers.

Average mass of elements in a plug-in hybrid vehicle



Source: <https://cen.acs.org/materials/chemical-elements-does-take-build/99/web/2021/07>

APSC 131 Course Outline



Rachel Baker
Chemical Engineering



Scott Parent
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Chemical Engineering



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Mechanical & Materials
Engineering



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Unit 1

Structure and Properties
of Atoms and Molecules

Unit 2

States of Chemical
Systems

Unit 3

Structure and Properties
of Aggregated Materials

Anatomy of an APSC 131 Lecture: Learning Plan

Keywords

- Chemistry of Materials
- Instructor Introduction
- Course structure and expectations

Resources

- APSC 131 OnQ Page
- Course Syllabus

Learning Outcomes

- Identify course topics and instruction team members
- Locate course resources and information
- Send emails with appropriate formats

Anatomy of an APSC 131 Lecture: Lecture Content

A complete set of APSC 131 lecture notes is comprised of:

1. Powerpoint slide decks posted to OnQ
 - Preparation: detailed information to be reviewed before class
 - Class: **theory** summary and **problems** to be presented by lecturers
2. Board work
 - Written solutions to questions related to the lecture learning outcomes.
3. Information provided orally by the professor
 - Tips and best practices based on the professor's experience.

A OneNote packet is provided in OnQ to assist you with notetaking.

Anatomy of an APSC 131 Lecture: Knowledge Development Questions

Knowledge Development Questions (KDQs) are:

1. The final slide of each lecture
2. Represent the key concepts you should learn from lecture
3. Modelled after quiz and exam questions
4. **The key to succeeding in this course**

Solutions posted at the end of each week.

Direct questions to your TAs or send an email (apsc131instructor@queensu.ca).

Tutorials

Why are we here?

- Practice concepts introduced in lecture
- Clarify difficult course material with knowledgeable personnel
- Problem solving sessions with weekly exercises

Who will guide you?

- One of our qualified Teaching Assistants

What will you learn?

- How to approach engineering problems for exams
- Advice on your future as an engineering student at Queen's

Deliverables – Homework, Coursework, and Exams

Weekly Homework Assignments – Best 8 of 10 (8%)

- Weekly OnQ Quizzes (approx. 30-minute commitment)
- Marked based on effort and completeness (out of 4 marks)

Weekly Tutorial Exercises – Best 8 of 10 (16%)

- Weekly during tutorial (approx. 50-minute commitment)
- Marked based on effort and completeness (out of 4 marks)

Midterm – Week 8 (26 %)

- November 5, 2026 at 6 PM (3-hour duration) – Rooms to be announced

Final Exam – Week 8 (50 %)

- Date and Time (TBD)

Allowed resources: Casio 991 calculator, writing utensil, straight edge

Equation sheets will be provided for assessments. Posted in OnQ.

Assessments – Remark Requests

ALL REMARK REQUESTS MUST BE SUBMITTED VIA THE COURSE FORM

(Form on OnQ -> Content -> BEGIN HERE -> Informal Remark Request Form)

Must provide a justification for specific marks to be reviewed.

Questions about the submission process can be directed to:

E-mail – APSC131instructor@queensu.ca

Academic Integrity

Engineers have a duty to:

- Act with devotion to the high ideals of personal honour and professional integrity
- Give proper credit for engineering work

Departures from academic integrity are antithetical to the development of an academic community at Queen's. Departures include:

- Plagiarism, forgery, copying from or impersonating another student
- Use of unauthorized materials or unauthorized collaboration
- Unauthorized sharing of course materials online or with other students

Possible Sanctions:

- Warning, loss of grades, failure of the course, withdrawal from the University

<https://engineering.queensu.ca/policy/academic-integrity.html>

Course Expectations

Expectations of you:

- Review lecture preparation slides
- Attend lecture and tutorial
- Review lecture notes and practice Knowledge Development Questions
- Stay curious and ask questions of the instruction team

Expectation of instruction team (Instructors, Coordinator, TAs):

- Deliver course content: lecture and tutorials
- Answer student questions
- Assess student mastery of course material
- Give feedback in a timely manner
- Share our knowledge of engineering and chemistry

Course Resources

OnQ: Contains all information about the course and university resources

- Each lecture contains a list of additional learning resources on the intro slide

Optional Textbook:

General Chemistry: Principles and Modern Applications, 11th edition

By: Ralph H. Petrucci, F. Geoffrey Herring, Carey Bissonnette

Publisher: Pearson Canada

Required Calculator:

Casio 991 (only)

Course Resources

Have Questions? Send an email:

E-mail – APSC131instructor@queensu.ca

The subject must include one of the following:

LECTURE Section xxx – for lecture questions

TUTORIAL Section xxx – for tutorial questions

HOMEWORK – for homework questions

EXAM – for exam questions

GENERAL – for questions not specific to lecture, tutorial, homework or exams

Examples: APSC 132 Tutorial Section 201 – Week 3 Problem Set Question
 APSC 132 Midterm Exam – Location and Time

If your email does not have a clear subject, response time may be longer.

Course Resources

Have Questions? Send an email:

E-mail – APSC131instructor@queensu.ca

Your signature block must include:

Name

Student ID Number

Example: Jonny Appleseed
9xx1xx9x

If your email does not have your name and ID #, response time may be longer.

Accommodations

Check Ventus for information on assessments. Locations and times may differ from the posted times for the course.

Contact QSAS for information on your accommodations and to get advice on accessing accommodations.

Wellness and Mental Health Resources for Students

Eng Resources	Queen's Resources	Other Resources
<u>Wellness Advisor</u> Unsure of the type of support and resources you need? Our Wellness Navigator can determine your individual needs and connect you with the appropriate resources and support.	<u>Queen's Student Wellness Services</u> Medical, mental health, accessibility and health lifestyle appointments. Also offers crisis counselling and a new walk-in counselling clinic (Mon-Fri. 1-4 p.m.). Phone: 613-533-2506 or Visit: in person 1st Floor Mitchell Hall	<u>Empower Me</u> Empower Me provides Queen's students with virtual access to therapy, coaching and 24/7 crisis support. Access through the <u>Dialogue App</u>
<u>Program Advisor, Accommodations/Considerations (absences)</u> Email: <u>engineering.aac@queensu.ca</u>	<u>Peer Support Centre</u> The Peer Support Centre offers drop-in hours for students to speak with supportive and compassionate student volunteers.	<u>GOOD2TALK</u> text GOOD2TALKON to 686868 to chat with a trained volunteer Crisis Responder. The service is free, confidential and available 24/7.
<u>Embedded Counsellor</u> Are you looking for short-term counselling to work on personal issues? Book in with our Embedded Counsellor.	<u>Faith and Spiritual Life</u> Offers multi-faith, non-judgmental support for religious, spiritual, personal, and financial problems, concerns or crises.	<u>BIPoC Counsellor</u> Saranya has joined the Smith Engineering Student Services team to provide additional counselling services to our BIPoC (Black, Indigenous, and People of Colour) Smith Engineering students.

To Do List...

1) Put important dates in your calendar

- Lecture and tutorial times
- Midterm exam

2) Access OnQ page

- Find syllabus, lectures, tutorials, and practice problem sets

3) Make a study plan (Time blocking)

- Lecture preparation
- Lecture notes review (for 10 min. on same day sometime before bed)
- Knowledge Development Questions and Tutorial Preparation

Knowledge Development Questions

Reflecting on Materials

1. **Identify** three materials that you are most interested in. **Report** how these materials are used in engineering.
2. **Report** where to contact the instruction team.
3. **Identify** a plan for success in this course (and first-year generally).

Welcome to APSC 131!



Rachel Baker
Chemical Engineering



Scott Parent
Chemical Engineering



Hoang Dang
Chemical Engineering



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Mechanical & Materials
Engineering



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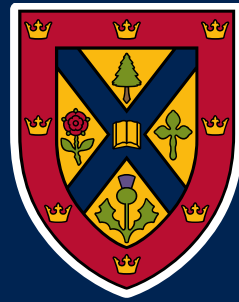
E-mail: APSC131instructor@queensu.ca
Office hours: By appointment

References

- Ice image: <https://medium.com/adventures-in-consumer-technology/critical-states-f293c17fa6d6>
- Breaking ice image: <https://evoke.pro/articles/63/>
- Shaved ice image: DOI: [10.3389/fpsyg.2021.638703](https://doi.org/10.3389/fpsyg.2021.638703)

Resources

- Email: APSC131instructor@queensu.ca
- APSC 131 OnQ page



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